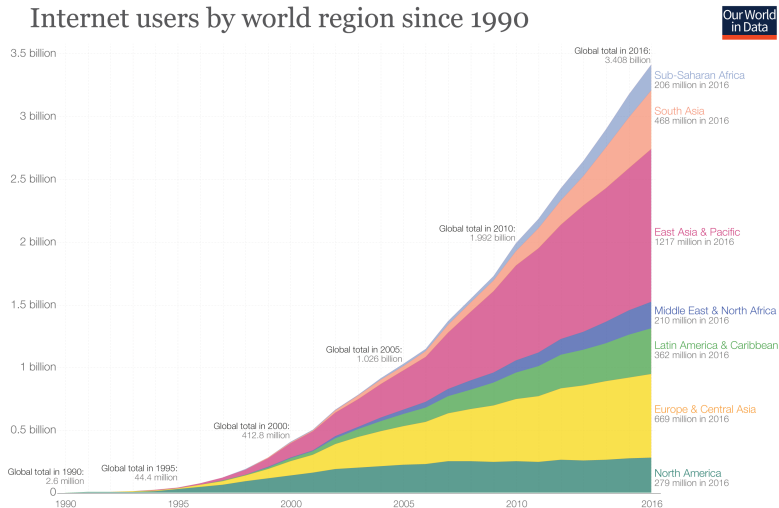


Advanced Data Analytics — Wrap-up

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December 15th, 2025 | 10:15 - 12:00: Internef 126 | 16:30 - 18:00: Anthropole 3185

Big Data and its availability



Data source: Based on data from the World Bank and data from the International Telecommunications Union. Internet users are people with access to the worldwide network. The interactive data visualization is available at OurWorldinData.org. There you find the raw data and more visualizations on this topic. Licensed under CC-BY-SA by the author Max Roser.

Why Machine Learning?

- ▶ **Machine learning** aims at **gaining insights from data** and making predictions based on it.
- ▶ **Build a model** that is **a good and useful approximation to the data**.
- ▶ Machine learning methods have been investigated for **more than 60 years**, but became mainstream only recently due to more data being available and advances in computing power ("**Moore's Law**").
- ▶ There is no need to "learn" to calculate, for example, the payroll.
- ▶ Learning is used when:
 - ▶ Human expertise does not exist (navigating on Mars).
 - ▶ Humans are unable to explain their expertise (speech recognition).
 - ▶ Solution changes in time (routing on a computer network)
 - ▶ Solution needs to be adapted to particular cases (user biometrics).
- ▶ You're relying on machine learning every day, maybe without being aware of it! you certainly use a Smart Phone?

Set some terminology straight

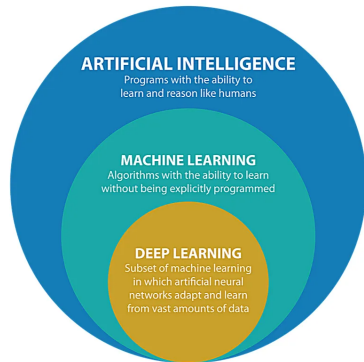
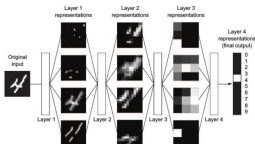
Artificial intelligence (AI)

Can computers be made to “think”? — a question whose ramifications we’re still exploring today. A concise definition of the field would be as follows: the effort to automate intellectual tasks normally performed by humans.

Machine Learning (ML)



Deep Learning as a particular example of an ML technique



F. Chollet (2018) — ML with Python

ML Applications

- ▶ **Association.**

- ▶ **Supervised Learning.**

Assume that training data is available from which they can learn to predict a target feature based on other features (e.g., monthly rent based on area).

- ▶ **Classification.**

- ▶ **Regression.**

- ▶ **Unsupervised Learning**

Take a given dataset and aim at gaining insights by identifying patterns, e.g., by grouping similar data points.

- ▶ **Reinforcement Learning.**

Supervised Regression

- ▶ Regression aims at predicting a numerical target feature based on one or multiple other (numerical) features.
- ▶ Example: Price of a used car.
 - ▶ x : car attributes
 - ▶ y : price
 - ▶ $y = h(x|q)$
 - ▶ $h()$: model
 - ▶ q : parameters

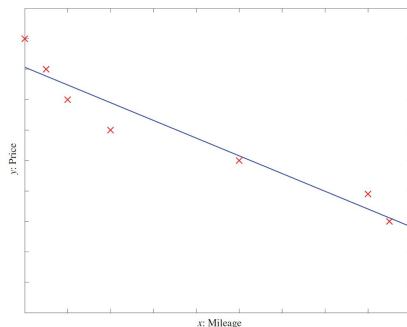
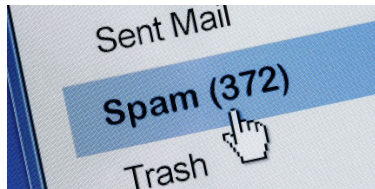


Fig. from Alpaydin (2014)

Supervised Classification

Example 1: Spam Classification

- ▶ Decide which emails are Spam and which are not.
- ▶ **Goal:** Use emails seen so far to produce a good prediction rule for **future** data.



Example 2: Credit Scoring

- ▶ Differentiating between low-risk and high-risk customers from their income and savings.
- ▶ **Discriminant:** IF $\text{income} > \theta_1$ AND $\text{savings} > \theta_2$ THEN low-risk ELSE high-risk

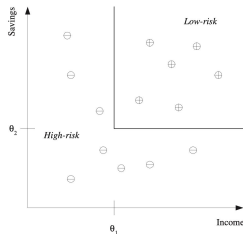
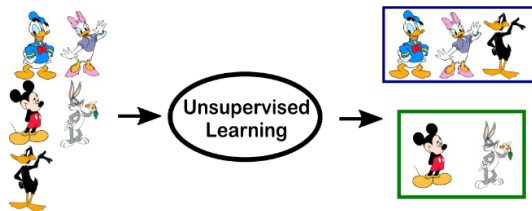
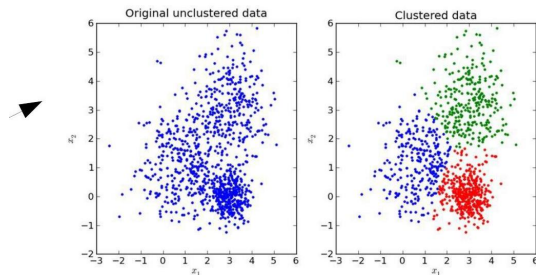


Fig. from Alpaydin (2014)

Unsupervised Learning

Learning “what normally happens”.

- ▶ No output.
- ▶ Clustering: Grouping similar instances.
- ▶ Example applications:
 - ▶ Customer segmentation.
 - ▶ Image compression: Color quantization.
 - ▶ Bioinformatics: Learning motifs.

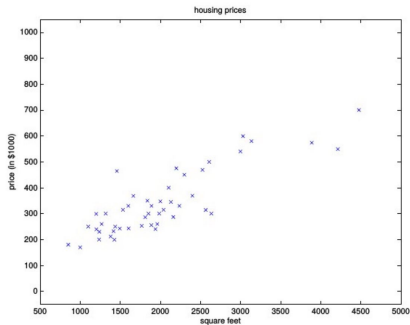


Reinforcement Learning

- ▶ Learning a policy: A sequence of outputs.
- ▶ No supervised output but delayed reward.
- ▶ Credit assignment problem.
- ▶ Game playing.
- ▶ Robot in a maze.
- ▶ Multiple agents, partial observability, ...

Building an ML Algorithm (II)

Living area (feet ²)	Price (1000\$s)
2104	400
1600	330
2400	369
1416	232
3000	540
⋮	⋮

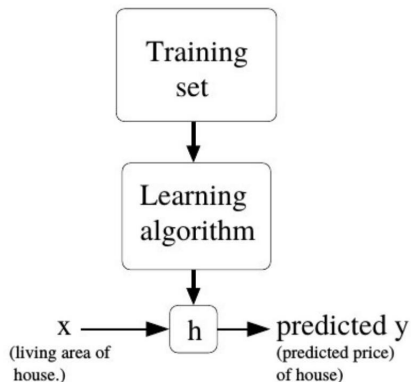


Given data like this, how can we learn to predict the prices of other houses as a function of the size of their living areas?

Building an ML Algorithm (III)

- ▶ $x(i)$: “**input**” variables
(living area in this example),
also called **input features**
- ▶ $y(i)$: “**output**” / **target variable** that we are trying to predict (price).
- ▶ **Training example**: a pair $(x(i), y(i))$.
- ▶ **Training set**: a list of m training examples $(x(i), y(i)); i = 1, \dots, m$

To perform supervised learning, we must decide how we're going to represent **functions/hypotheses** h in a computer.



Building an ML Algorithm (IV)

► Model/Hypothesis:

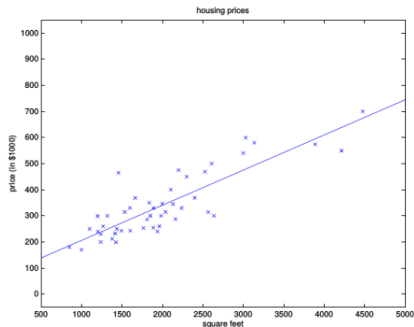
$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

θ_i 's: parameters

► Cost Function:

$$J(\theta) = \frac{1}{2} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

⇒ Minimize $J(\theta)$ in order to obtain the coefficients θ .



What we covered

1. Introduction to Machine Learning, and a crash course on Programming in Python
2. Mathematics for Machine Learning (Linear Algebra, Probability, and Optimization)
3. Supervised Learning - the general idea, Linear Regression (with multiple variables), Gradient Descent, Polynomial Regression, Tuning Model Complexity, Introduction to Pandas
4. k-Nearest-Neighbours, Naive Bayes, Decision Trees, Combining Models (Boosting, Bagging,...), Kernels
5. Gaussian Process Regression

What we covered

1. Bayesian Active Learning, Gaussian Process Classification
2. Neural Nets (Feed-forward Networks, Network Training, Error Back-propagation, Some Notes on Over-fitting)
3. Recurrent Neural Networks, LSTM
4. Transformers, NLP
5. k-Means, Gaussian Mixture Models, EM-Algo. Hierarchical Clustering, Density-based Clustering
6. The curse of Dimensionality, Active Subspaces & GPs, Principal Component Analysis
7. Introduction to Reinforcement learning

Data Scientist (n.):

Person who is better at statistics than any software engineer and better at software engineering than any statistician.

Data Science

The activities of Greater Data Science are classified into 6 divisions:

1. Data Exploration and Preparation
2. Data Representation and Transformation
3. Computing with Data
4. Data Modeling
5. Data Visualization and Presentation
6. *Science about Data Science



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AND**

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